

Problem species and numerical artifacts: a compendium

This document collects the numerically difficult species and the physical (as opposed to purely infrastructural) artifacts encountered while training and evaluating neural exchange-correlation functionals in this repository, for use in the paper’s methods and appendix material. Each section states the symptom with its magnitude, the root cause with the governing relations, the detection signal that now fires on a recurrence, the remedy the pipeline applies, and the status. Provenance is given inline in a compact form: `HISTORY <date>` denotes the dated entry in `xcquinox/alec/HISTORY.md`; all other paths are repository-relative and name the log, docstring, or analysis document in which the quoted number was measured or is pinned. Production identity, unless stated otherwise, means 6-311++G(3df,2pd), grid level 3, density fitting on, orientation-lock strength 3×10^{-5} .

1. C2: a two-configuration SCF landscape

Symptom. The C2 dimer is the only species of the 214-species benchmark pool whose plain PBE kernel fails to converge (HISTORY 2026-07-26). At the held-out evaluation identity (RKS/PBE, 6-311++G(3df,2pd), grid 3, lock 3×10^{-5}) DIIS oscillates for all 100 cycles between two SCF configurations with a trajectory energy spread of 1.204×10^{-1} Ha, 73 of 100 cycles in the lower basin, ending at an arbitrary phase of the oscillation (final cycle $E = -75.7175899504$) (scratch/v6_diag/repro_c2_pbe_branch.log).

Root cause. Two distinct SCF solutions exist, consistent with the multireference character of the C2 ground state: a second-order (SOSCF) solve converges either to $E = -75.8167407121$ Ha (internally stable; externally unstable toward a UKS symmetry breaking, as is the higher solution) or to $E = -75.7368945310$ Ha (internally unstable), split by 7.984×10^{-2} Ha = 50.10 kcal/mol (scratch/v6_diag/repro_c2_pbe_branch.log).

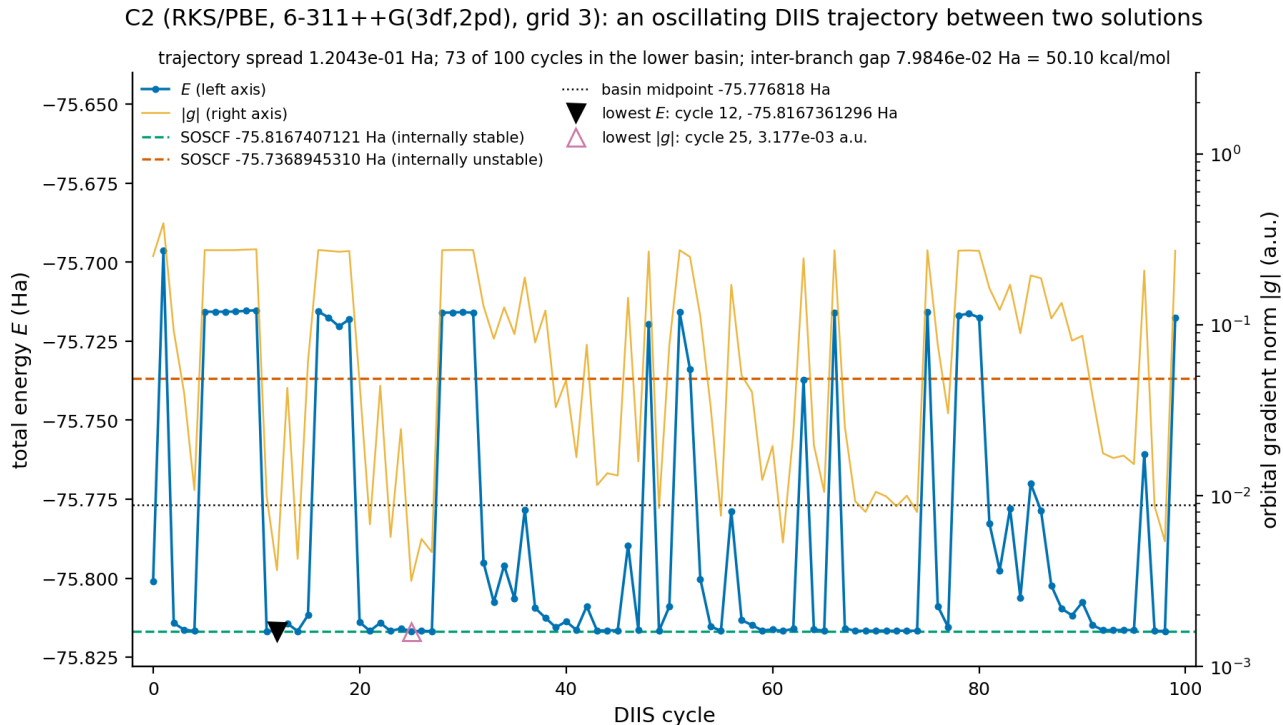


Figure 1: C2 at the held-out evaluation identity (RKS/PBE, 6-311++G(3df,2pd), grid 3): the 100-cycle DIIS trajectory oscillating between the two converged SCF solutions, with the lowest-energy (cycle 12) and lowest-gradient (cycle 25) trajectory points marked.

The left axis carries the total energy of each DIIS cycle and the logarithmic right axis the orbital gradient norm; the dashed horizontals are the two converged second-order solutions, -75.8167407121 Ha (internally stable) and -75.7368945310 Ha (internally unstable), and the dotted horizontal is their midpoint at -75.776818 Ha, taken as

the dividing line for counting rather than as a physical separatrix. The trajectory crosses that line repeatedly and never settles: 73 of the 100 cycles lie below it, the energies span 1.2043×10^{-1} Ha, and the phase the cycle cap happens to fall on – here the upper branch – carries no information about which solution the species has. Both markers lie on the lower branch: the lowest-energy point at cycle 12, -75.8167361296 Ha, sitting 4.58×10^{-6} Ha above this draw’s converged solution – the same separation the acceptance check under Detection reports as -4.09×10^{-6} Ha on a different draw, that check signing the excess as converged minus trajectory minimum, so the two figures differ in sign convention and not in direction – and the lowest-gradient point at cycle 25, where $|g|$ falls to 3.177×10^{-3} a.u.; these are the two seeds the remedy below distinguishes, the first for the orbital-pair rerun and the second for the second-order stage. What the figure establishes is the landscape the next paragraph’s aufbau flip rides on: a rescue must take its seed from a trajectory that visits both basins, and the quality of that seed does not by itself decide which one is reached.

The dm0-ingestion aufbau flip. Which solution a density-seeded second-order solve reaches is not controlled by the quality of the seed: PySCF’s SOSCF ingests `dm0` by diagonalizing $F[\text{dm0}]$ and re-occupying by aufbau, and the ground solution of C2 is non-aufbau in its own Fock. Seeding SOSCF with the trajectory’s lowest-energy density ($E = -75.8167361$) converges 0.0798 Ha uphill onto the internally unstable higher solution, and even the converged ground solution’s own exact density flips to the higher branch ($E = -75.7368945380$); seeding from the same point’s orbital pair instead converges to the ground solution in 2 macro-iterations (xcquinox/alec/data.py, `_converge_reference_scf` docstring; scratch/v6_diag/repro_c2_pbe_mo_start.log). Which branch a density-seeded start lands on is draw-dependent – the unconverged oscillation is chaotic; local draws stayed on the ground branch while 7 cluster evaluations stamped the higher one (data.py docstring).

Detection. Two independent gates. (i) At reference generation, a converged solution – a second-stage rescue and a DIIS-converged endpoint alike – is accepted only within a branch tolerance of 10^{-4} Ha of the DIIS trajectory’s own minimum energy. The measured same-basin excesses at equilibrium identities are -2.97×10^{-7} Ha (Li/SCAN), -4.09×10^{-6} Ha (C2/PBE), $+8.26 \times 10^{-7}$ Ha (O/PBE rescue) and $+8.38 \times 10^{-6}$ Ha (S/SCAN, a DIIS-converged endpoint over its own trajectory minimum), so the tolerance sits roughly 12 $\times$ above the largest of them and nearly three decades below the measured inter-branch gap of 7.984×10^{-2} Ha; a sweep of every species in the current data-generation pools (28 atom runs plus the rescue identities) found none above $+8.4 \times 10^{-6}$ Ha (data.py, `_REFERENCE_SCF_BRANCH_TOL`). (ii) At the figure layer, any species whose model-free E_{PBE} spreads across specs by more than 10^{-4} Ha (SCF reconvergence noise is $\sim 2.5 \times 10^{-6}$ Ha) is excluded from every pooled reference baseline with a printed warning naming the species and values (HISTORY 2026-08-12).

Remedy. The reference SCF runs DIIS first, then a second-order stage seeded from the lowest-gradient point of the DIIS trajectory; a converged solution exceeding the branch tolerance – from either stage – is rerun from the minimum-energy point’s recorded orbital pair (immune to the aufbau re-occupation), the lower converged solution is kept, and a persisting excess is refused with the refusal naming the rerun’s own outcome. The residual exposure of the acceptance check is a false refusal, never a wrong recorded value: at a near-degenerate geometry (C2 stretched to $r = 1.60$ Å) the rescue’s converged endpoint is itself reproducible only to 1.31×10^{-4} Ha across draws, so such a species can refuse on an unlucky draw (data.py, `_REFERENCE_SCF_BRANCH_TOL`). The production path converges equilibrium C2 in 105 total cycles (`diis+newton`) to $E = -75.8167407121$ with a stamped orbital gradient of $1.8\text{--}2.2 \times 10^{-6}$ (scratch/v6_diag/verify_c2_branch_fix.log).

Earlier episodes.

- *Reference-grid drift* (HISTORY 2026-07-26). PySCF density-prunes the DFT grid once, at the first SCF cycle, on the density held at that moment (`small_rho_cutoff = 10^{-7}`). The tiered convergence escalation rebuilt a fresh mean field seeded `dm0=make_rdm1()`, whose grid was pruned on the non-converged 50-cycle DIIS density; the stored reference grid therefore depended on the generation-time trajectory. C2’s reference carried 26840 points (26840/26848/26856 across DF/lock configurations, basin-sensitive between the two solutions, $E = -75.8168$ against -75.7158 for the escalated draw) against the 26568 points every consumer computes from the guess-pruned grid; the eval-side loader’s shape gate then dropped the species from every completed eval. Controls: N2 (26616) and F2 (26568) converge on the plain kernel and match exactly. Remedy: the stored grid is pinned to the FIRST kernel attempt regardless of which tier converges, and the density-fitting identity is stamped into the cache (`density_fit_used`); deleting the final `.npz` alone re-drifts, so the intermediates must be deleted with it. The regenerated reference reads 26568 points (probe: hpcjobs/dfs6311_c2_ref_probe.py).
- *Cross-arm reference drift* (HISTORY 2026-08-12, 2026-08-17). The same class resurfaced across two

concurrently evaluated sweep arms: $E_{\text{PBE}}(\text{C2}) = -75.816711949$ Ha in one arm against -75.757329256 Ha in the other – a 37.26 kcal/mol disagreement against $\leq 3.6 \times 10^{-12}$ Ha spread within each arm – and, within one arm, a single degraded evaluation whose 2-thread retry re-converged the C2 PBE SCF to a third value (-75.781328 , 24 mHa off). The cross-spec-spread guard excludes the species from pooled baselines; no beats-baseline verdict moved.

- *Benchmark gate refusal* (HISTORY 2026-08-18). During a local recomputation of transiently lost per-species energies, every patch was gated on the locally converged PBE energy matching the recorded one within 10^{-7} Ha (two decades above the $\sim 10^{-9}$ conv-tol reproducibility bound, five below the multi-solution signal). The gate fired exactly once: C2 converged 5.93×10^{-2} Ha away from the recorded solution and the patch was refused rather than computed from the wrong density.

Status. Closed at the reference-generation layer (branch-stable rescue, pinned grid). The only C2 benchmark reaction (`w411_c2_atomization`) sits in the STRICT held-out slice of every completed v6 cell (its `in_sample_overlap` empty in all 116 `per_reaction.json` records – the 29 completed specs, four evaluation channels each); the seven branch-affected evaluations were repaired in place by a gated patch tool rather than re-evaluated wholesale, in two rounds (the standard channels, then the best-loss channel after its checkpoint fetch), after which the audit reads 116 channels, 0 wrong, the cross-spec reference guard is silent and C2 rejoins the pooled PBE baselines on all four evaluation channels; the two cells completed after the repaired reference layer was deployed to the cluster arrived with the correct branch natively, with no patching (HISTORY 2026-09-01); the beats-baseline verdicts are unchanged with and without the species. In the earlier (v4-era) split the reaction sat in the validation slice. The drifted GGA-arm evaluation reference is scheduled for regeneration at array drain (xcquinox/alec/DEFERRED_WORK.md item 16).

2. Li: reference-SCF fragility of a diffuse-basis alkali

Symptom. The v6 meta-GGA data-generation stage refused on the Li atom: the SCAN reference SCF at 6-311++G(3df,2pd)/grid 3 under the 3×10^{-5} lock reaches the solution basin early (cycle 5: $E = -7.478697644723$, $|g| = 7.5 \times 10^{-4}$), the DIIS extrapolation then throws the density into an unphysical state at cycle 7 ($E \approx -5.04$) and wanders chaotically to the 100-cycle cap (endpoint -4.07 in one draw, -3.85 in another; the pre-explosion basin numbers reproduce across draws) (HISTORY 2026-08-30; `scratch/v6_diag/repro_li_scan.log`).

Root cause. A DIIS explosion after the basin is reached leaves the trajectory’s final point as the worst possible start for a rescue: a second-order solve seeded from the endpoint runs all 50 macro-iterations unconverged (stall gradients 1.2×10^{-3} to 4×10^{-3} across draws). The failure requires SCAN together with the diffuse ++ set – PBE converges in 5 cycles at the same identity, SCAN at def2-svp in 7, and Na (the protocol’s other alkali) in 7 at baseline. Level shifts (0.25 / 0.5), damping (0.5), a finer grid (level 5), and removing the lock were each measured and none converges it (`scratch/v6_diag/repro_li_scan.log`, `scratch/v6_diag/repro_li_scan_fixes.log`).

Remedy and status. The second-order stage now starts from the lowest-gradient density the DIIS trajectory visited, recorded by a callback: from that point SOSCF converges Li in 2 macro-iterations to $E = -7.4786979415$; the production path stamps `diis+newton`, 102 total cycles, gradient 5.97×10^{-6} against the $\sqrt{10^{-9}} = 3.16 \times 10^{-5}$ criterion (`scratch/v6_diag/verify_li_fix.log`). The landed value was checked independently: SOSCF from the minao, atom, and huckel guesses agrees to a 3.8×10^{-11} Ha spread, with an aufbau occupation, $S^2 = 0.750004$, internally stable; DIIS-converged records are bitwise unchanged, and of 48 production-setting rescues 45 are no-ops and the 3 that differ improve (HISTORY 2026-08-30). With the rescue in place the full polarized SCAN species set at the production identity generated cleanly (datagen wall 6:00; HISTORY 2026-08-31).

Earlier episode: poisoned-dm0 escalation (HISTORY 2026-07-05, Phase 19). The same atom failed differently in the SCAN baseline generator: plain DIIS diverged to a garbage density ($E \sim -4.9$ against the true -7.478), and the tiered escalation then seeded every tier with that garbage as `dm0`, so a `dm0`-seeded second-order solve inherited it and stalled – while the same solver from a fresh minao guess converged cleanly ($E = -7.4785$). The escalation gained fresh-guess tiers that ignore `dm0`; grid refinement was shown irrelevant (grid 3/4 identical).

3. OH, CH, NO: orientation degeneracy of $^2\Pi$ radicals

Symptom. For an orbitally degenerate $^2\Pi$ ground state, the singly occupied π hole can occupy either degenerate component, so the single-determinant density on a fixed grid is orientation-arbitrary while the energy is invariant. Measured on OH (UKS/PBE), the model-free density RMSE against a fixed CCSD reference across fresh processes read 9.18×10^{-4} , 2.74×10^{-3} , 2.66×10^{-3} , 1.94×10^{-3} – each reproducible within its process, all different across processes, all with correct $\int \rho = 9$ and $n_{occ}^{\alpha/\beta} = 5/4$; with OMP_NUM_THREADS=1 the value is reproducible across processes (2.333×10^{-3} twice). The scramble is threaded-BLAS non-associativity tipping the near-degenerate SCF between components (HISTORY 2026-07-01). At production scale the pooled def2-svp record shows the model-free PBE density error bit-identical across every spec for all 20 closed-shell species and varying for exactly three – HO 22.96 $\times$, CH 20.06 $\times$, NO 7.99 $\times$ (max/min over 208/114/50 evaluations) – at essentially constant energy (PBE total-energy spans of 1.1×10^{-6} , 8.2×10^{-8} , 4.4×10^{-8} Ha) (notebooks/analysis/DENSITY_DIAGNOSIS.md Sec. 3).

Consequence for training. With CH and NO in the training set, the density channel of the loss was owned by an error no functional can close. The measure throughout is the reference publication’s per-electron density error, $\epsilon_n = N_e^{-1} \int |\rho - \rho_{ref}|$, with N_e the quadrature integral of the reference density (Dick and Fernandez-Serra, Phys. Rev. B 104, L161109 (2021), eq. (20); xcquinox/alec/evaluation.py). CH carries an Eq.-20 per-electron density error of 1.55×10^{-1} against its stored reference (NO 7.78×10^{-2}) where the population median is 8.4×10^{-3} , and model-free PBE reproduces CH’s error to 0.4% (NN/PBE = 1.004); the two species own 68–98% of the channel wherever present, and the channel is flat across 200 epochs (median first-to-last ratio 0.998) while the energy channels fall by factors up to 2×10^5 (DENSITY_DIAGNOSIS.md; HISTORY 2026-08-03). 4 of 114 CH draws land at or below the closed-shell population median (smallest 0.61 $\times$), which a genuine functional-versus-reference gap cannot produce (a fixed functional against a fixed reference is one number) and component mismatch does.

Detection. The discriminating measurement is a MODEL-FREE error that varies across otherwise identical evaluations at constant energy: same energy, different density is a degenerate manifold. A cross-model or cross-spec spread of the PBE density reference above 2–5% relative now warns on the figures and excludes the species from pooled density anchors (HISTORY 2026-07-01, 2026-08-13). The degenerate-state class extends beyond the trio on the pre-lock def2-svp record: 12 of 198 species’ PBE density references varied across spec evaluations, led by the BH76 transition state RKT14 and bn($^3\Pi$)/ClO/NO/OF/OH/CH (HISTORY 2026-07-29).

Remedy: the orientation lock. A small, fixed, deterministic, traceless anisotropic-quadrupole operator $M = \sum_{ij} W_{ij} \langle \chi_\mu | r_i r_j | \chi_\nu \rangle$ (W symmetric, $\text{Tr } W = 0$, three distinct eigenvalues, generic principal axes, built about the nuclear-charge centroid) is added to h_{core} as λM identically in every path that produces a density for the degenerate radicals: the CCSD/HF reference SCFs, the PBE seed, and both differentiable-SCF backends (HISTORY 2026-07-02; xcquinox/alec/orientation_lock.py). Tracelessness is load-bearing: the first-order shift $\lambda \text{Tr}(MD)$ is ~ 0 for a near-isotropic density, so the lock splits the degenerate π pair without materially shifting energies. The calibrated strength $\lambda = 3 \times 10^{-5}$ induces a π splitting of $\sim 10^{-6}$ – 10^{-5} Ha, about four orders above float64/BLAS noise, while shifting a closed-shell PBE total energy by < 0.1 kcal/mol (HISTORY 2026-07-02; orientation_lock.py header).

Cache-identity episodes. The lock is only as good as its reach. Three successive gaps put locked SCFs against unlocked references for exactly these radicals: the training-reference generation path omitted the strength (HISTORY 2026-07-11); the final per-species reference cache was lock-blind – keyed only on (name, basis) – and silently reused unlocked files (HISTORY 2026-07-14); and CH and NO were confirmed unstamped mid-campaign (file dates predating the cache fix) and were relocked mid-run with the spec partition recorded (before/after boundary specs 0000-0023 / 0024-0087) (HISTORY 2026-08-03). The cache-currency predicates now compare the recorded lock against the requested identity, a legacy file that records none reading as 0.0 – a hit for an unlocked run and a miss for a locked one, in either direction (HISTORY 2026-08-24; external_refs.py:1748-1752).

Residual physics. The lock delivered reproducibility, not agreement: under it the production run has zero cross-spec scatter, but only HO landed on the component its stored reference used (1.36×10^{-4} , ordinary), while CH (1.58×10^{-3}) and NO (2.57×10^{-3}) sit on a different member of the manifold than their references (DENSITY_DIAGNOSIS.md Sec. 3). The reference publication met the same wall on the same class and scaled its density-loss weight λ_n by 0.01 for CH and OH; the corresponding per-species weighting here targets CH and NO (in this pipeline OH is the member that landed correctly), and is recorded as the ranked remedy rather than applied mid-campaign (HISTORY 2026-08-03). Related: the open-shell radicals HO, CN and NO (HO and NO

$^2\Pi$; CN $X^2\Sigma^+$) also basin-hop in the Wu-Yang OEP inversion used for V_{xc} references, handled with an inner-SCF level shift and a looser UKS tolerance (10^{-2} against 2×10^{-3} RKS) (HISTORY 2026-05-02, Phase 5).

Status. Lock productionized across every density-producing path and stamped into every cache identity; component agreement (as opposed to reproducibility) remains an open, documented limitation with the per-species weighting as the recorded remedy.

4. Open-shell free atoms (O, N, F): degenerate p-shells and exact spin scaling

Free atoms anchor every atomization energy in the training pools, so free-atom defects enter every reported AE.

Two-block spin-scaling defect. Exchange spin scaling

$$E_x[n_\alpha, n_\beta] = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta])$$

(Oliver and Perdew, Phys. Rev. A 20, 397 (1979)) is exact only when each channel is evaluated on the spin-unpolarized system it names. The production open-shell exchange passed the TOTAL-density iso-orbital indicator unchanged into both channels, making the open-shell functional a different functional from the closed-shell one: with libxc SCAN in place of the network, the superseded two-block evaluation errs by -47.9 mHa (-30.1 kcal/mol) on the O atom, -40.5 mHa on N, -26.8 mHa on OH at fixed density, and 48.5 mHa (30.5 kcal/mol) self-consistently on O; the pretrained meta-GGA consequently over-bound H₂O/N₂/CH₄ by $30.5/55.9/20.8$ kcal/mol relative to SCAN, of which transforming the indicator alone recovers 63–86% (residual offsets $-7.6/-7.9/-7.6$ kcal/mol; fractions recomputed from those offsets, the smallest on CH₄ at 63.46%) (HISTORY 2026-08-20, Phase 39; 2026-08-23, Phase 43; notebooks/analysis/NOTES_v5_mgga_vs_scan.md). The remedy computes every density-matrix-derived descriptor of channel σ on the symmetric doubled density $\text{diag}(P_\sigma, P_\sigma)$ – for the indicator, $\alpha_\sigma = \alpha(2\rho_\sigma, 4\sigma_{\sigma\sigma}, 2\tau_\sigma)$ – with correlation kept on the total density and ζ ; the three-block energy reproduces PySCF’s spin-polarized (`spin=1`) SCAN exchange to 1.8×10^{-15} Ha on O and OH (the GGA path reproduces `spin=1` PBE to the same 1.8×10^{-15} Ha), and the assembled potential is the finite-difference derivative of that energy to 1.0×10^{-10} Ha worst case (HISTORY Phase 43). The discriminating oracle was one libxc `spin=1` call; a re-derivation sharing the code’s frozen-feature assumption would have passed.

Orientation dependence of open p-shells. An open p shell holding 1, 2, 4, or 5 electrons is a P term whose hole converges to an arbitrary orientation; the exact functional is orientation-invariant but a quadrature on a fixed grid is not, and the meta-GGA feels it: the SCAN E_{xc} of the free O atom spreads by order 0.1 mHa between independent unconstrained SCFs at def2-svp/grid 3 (0.26 mHa over one triple of runs, 0.084 over another; 0.21 mHa for F at sto-3g/grid 1) against 1.6×10^{-3} mHa for PBE – a meaningful fraction of the 1.0 mHa certificate tolerance, decided by which orientation the SCF happened to reach. The fidelity certificate named here and below is the per-architecture pretraining gate: each pretrained network’s E_{xc} is evaluated on the parent functional’s own self-consistent densities for the free atoms of the BH76/W4-11 pools, the molecular differences are folded into atomization-energy offsets, and the architecture passes only when the worst free-atom error is at most 1.0 mHa and the worst atomization offset at most 1.0 kcal/mol (xcquinox/alec/cluster/fidelity.py). Certificate reference densities of such atoms are therefore built under the orientation lock, after which two independent locked runs agree to 3.4×10^{-11} mHa; spherical atoms are exempt (HISTORY 2026-08-23, Phase 40). The pretraining-data generator likewise refuses a spatially degenerate open-shell atom below grid level 3 or at lock 0 without a recorded waiver: at grid level 1 four locked draws of the O atom spread 3×10^{-3} in ρ , 0.64 in the iso-orbital indicator, and 3.7×10^{-6} Ha in the stored exchange energy while agreeing to 9×10^{-10} Ha in total energy (grid 3 reproduces to 3×10^{-11}) – two materially different files could otherwise carry one manifest identity (HISTORY 2026-08-24, Phase 42).

Free-atom SCF stalls. The locked PBE O atom at def2-svp/grid 1 exhausted PySCF’s default 50-cycle cap in 2 of 3 attempts, and the unlocked SCAN O atom stalled in 2 of 12 processes from the minao guess (one at a final gradient 8.98×10^{-5} against the 3.2×10^{-5} criterion), rescued by the second-order stage in 106 total cycles (HISTORY Phase 40; data.py `_converge_reference_scf` docstring). Tightening the reference `conv_tol` from 10^{-9} to 10^{-10} was measured and rejected: at 10^{-10} two processes wander 37 and 42 cycles along the lock’s weakly broken flat direction and land on different densities (1.4×10^{-3} relative in ρ on 94% of the grid) – the lock’s reproducibility is calibrated at PySCF’s default (data.py, `_REFERENCE_SCF_CONV_TOL` comment). On the O atom

the two SCF backends, cross-seeded from the same PBE density, reach two members of the degenerate manifold 9.7×10^{-5} Ha apart – a property of the manifold, not a backend defect (HISTORY 2026-08-23, Phase 43).

5. The iso-orbital indicator in density tails

The meta-GGA rung enters through the SCAN/DFS iso-orbital indicator

$$\alpha = \frac{\tau - \tau_W}{\tau_{unif}}, \quad \tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{unif} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3},$$

whose division by $\tau_{unif} \propto n^{5/3}$ (and the resulting $d\alpha/d\sigma \sim n^{-8/3}$ sensitivity) concentrates every pathology in low-density regions.

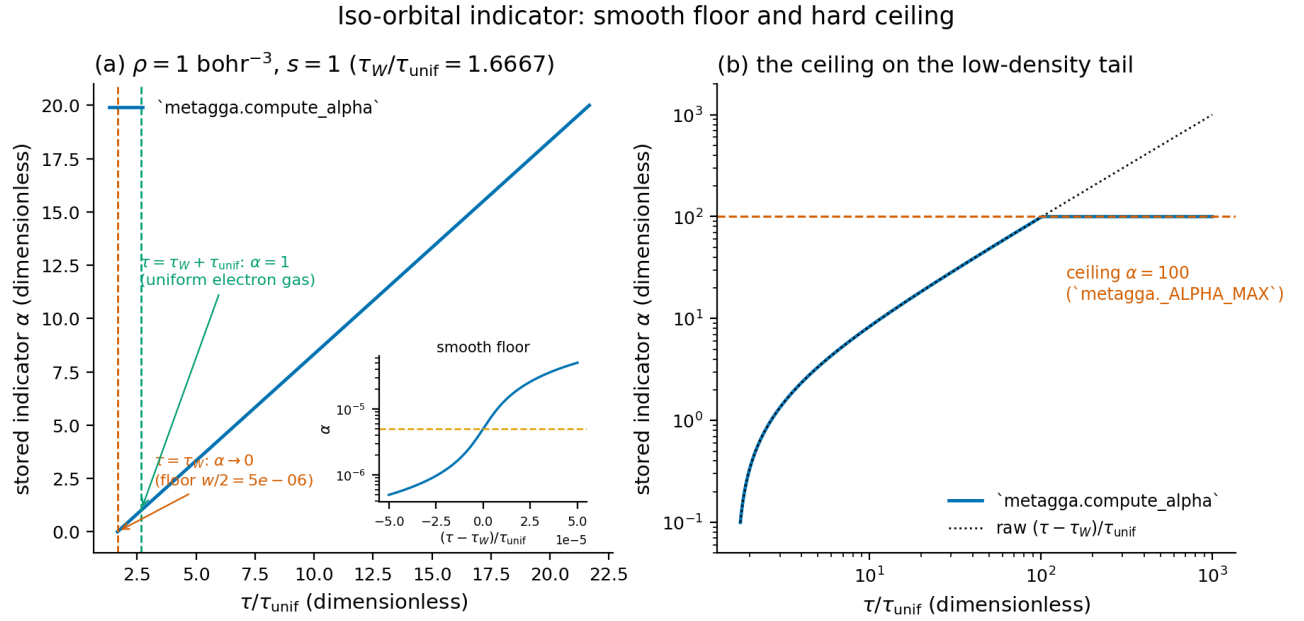


Figure 2: The stored iso-orbital indicator produced by `metagga.compute_alpha`, drawn against τ/τ_{unif} at unit density and reduced gradient $s = 1$: (a) the linear regime with the smooth floor inset; (b) logarithmic axes, on which the stored value departs from the raw ratio at the ceiling.

Panel (a) fixes the density and the reduced gradient so that the ratio τ_W/τ_{unif} is 1.6667, the abscissa at which the raw indicator vanishes; the stored indicator follows the raw ratio to within 5×10^{-6} across the whole panel and passes through unity at $\tau = \tau_W + \tau_{unif}$, the uniform-electron-gas point. The inset resolves the immediate vicinity of $\tau = \tau_W$ on a logarithmic ordinate: the stored value passes through the floor value $p(0) = \delta/2 = 5 \times 10^{-6}$ of Sec. 5.2 exactly at $\tau = \tau_W$ and continues smoothly to 4.95×10^{-7} at the panel's left edge, where the raw indicator is negative – off the SCF manifold, on which $\tau \geq \tau_W$ holds and the floor is the smallest stored value. Panel (b) repeats the map on logarithmic axes out to a raw ratio of 1000: stored and raw coincide until the raw value reaches 100, past which the stored curve is flat at the ceiling of Sec. 5.1 (`metagga._ALPHA_MAX` = 100), with 151 of that panel's 601 sampled points sitting on it and the largest raw value compressed tenfold. For the physical raw indicator ($\tau \geq \tau_W$) the two bounds give the domain of every indicator the network sees, $[5 \times 10^{-6}, 100]$, and they are what keeps the division by τ_{unif} finite in the tail.

5.1 Diffuse-tail ill-conditioning. On the diffuse ++ basis the Li 2s tail reaches $\rho = 5.4 \times 10^{-13}$ at grid 2; with only a value floor in place, α reached values $\sim 10^{28}$ (overflowing the forward SCF energy) and gradients $d\alpha/d\sigma \sim 10^{59}$, with the XC-kernel second derivative $\sim 4.2 \times 10^{43}$ at a point whose bare quadrature weight is 295, not zero – the “negligible because ~ 0 integration weight” argument holds for the energy (weight $w\rho$) and fails for the kernel the gradient uses (weight w). The signature was a meta-GGA training loss of NaN at step 0

on the `bh76:HLi` group; the GGA and rung-3.5 architectures, which divide by no density power, were immune (HISTORY 2026-07-04, Phase 17). The remedy caps the value ($\alpha \in [0, 100]$), above which the DFS/SCAN gate has saturated: $\tanh^2(\ln(101/2)) = 0.998$) with the same cap applied to precomputed and live values. A tail gradient freeze shipped at the same time was later found to be a defect in its own right – a potential that is not the derivative of its energy – and was removed once the feature-response term was assembled exactly; removing it takes $\max |d\alpha/d\sigma|$ from 1.15×10^{14} to 2.20×10^{31} on Li at the production basis, yet the full-SCF training gradient remains finite including at the original failing configuration (25 cycles, pretrained weights, H/Li/LiH), and the meta-GGA architectures land at the descriptor-free control’s finite-difference floor (2.1×10^{-10} RKS) (HISTORY 2026-08-06).

Measurement caveat banked with that change: the diffuse-basis H atom does not converge in 25 SCF cycles, and under multithreaded BLAS its unconverged energy and training-gradient magnitude wander between identical runs – four repeats spanned 60 mHa (38 kcal/mol) in energy and eight orders in $|g|_{\max}$ (5.7×10^{-2} to 1.5×10^7) – while `OMP_NUM_THREADS=1` is bit-identical across runs. A single run on an unconverged open-shell diffuse-basis system is not evidence of anything; such measurements are repeated or pinned to one BLAS thread (HISTORY 2026-08-06).

5.2 One-orbital spin channels (H α , Li β ; the H atom’s total density). A spin channel holding one electron is a single orbital, so $\tau = \tau_W$ identically on the SCF manifold and the raw indicator is the rounding residue of that cancellation – at most 6.6×10^{-10} on every grid point with $2\rho_\sigma > 10^{-8}$ across the three tested identities (xcquinox/alec/metagga.py, `_ALPHA_SMOOTHING_WIDTH` comment). A hard lower clip $\max(\alpha_{\text{raw}}, 0)$ made the derivative one-sided exactly on that manifold, and autodiff returned whichever side the rounding selected: the β -channel feature-response term of Li’s Fock matrix, itself of magnitude 1.13 Ha, moved by 0.93 Ha under a 10^{-14} relative change of the density matrix (H’s by 4.2×10^{-3} Ha) (metagga.py, `compute_alpha` docstring; HISTORY 2026-08-23, Phase 43). The cure replaces the clip with a C^∞ positive part

$$p_\delta(x) = \frac{x + \sqrt{x^2 + \delta^2}}{2}, \quad \alpha = \min(p_\delta(\alpha_{\text{raw}}), 100),$$

with width $\delta = 10^{-5}$ in indicator units (equivalently $10^{-5} \tau_{\text{unif}}$ in kinetic-energy-density units, so the construction is invariant under uniform density scaling); the smoothed indicator then ranges over $[\delta/2, 100] = [5 \times 10^{-6}, 100]$. The width is anchored to measurement, not chosen: it clears the largest on-domain rounding residue (draw-dependent, $1.3\text{--}3.7 \times 10^{-6}$ in the worst band) by $2.7\text{--}7.7\times$, and its energy cost – the H atom sits at the floor $\delta/2$ on every point – is $+1.17 \times 10^{-7}$ Ha of SCAN exchange (Li β : $+3.1 \times 10^{-7}$), linear in the width and 8.5×10^3 below the 1.0 mHa certificate tolerance (metagga.py; HISTORY 2026-08-24). Under the same 10^{-14} probe the H Fock now moves 3.6×10^{-12} Ha, and the H and Li fixed points reproduce across perturbed seeds to 0.0 and 1.9×10^{-14} Ha; the occupancy-keyed solver gate and the oracle’s straddle mask, each a place where the implementation and its check had agreed to look away from the same point, were retired. The indicator definition is recorded in the pretraining-data identity (`ALPHA_DEFINITION`), so a file computed under the clip is stale for a run under the smooth part. The smoothing’s own role is the one measured above – the one-orbital Fock response, cured from 0.93 Ha to 3.6×10^{-12} Ha – and it is not the source of the pretraining floor. SCAN-parent pretraining starts at a loss floor of 3×10^{-14} where PBE parents floor at 2.7×10^{-32} , and that floor belongs to the indicator’s CEILING: the anchored network inverts the stored smoothed column exactly before the parent reads it, so a stored $p(0) = 5 \times 10^{-6}$ recovers α_{raw} to round-off and the end-to-end anchored exchange loss on the committed mesh is 7.6179×10^{-32} ; the mesh-carrying and mesh-free floors stand in the ratio 0.7000000000000004, exactly the 0.7 atomic share of the loss weighting, so the mesh block contributes at most 6×10^{-29} ; and the H atom, one-orbital on every row, floors at 2.85×10^{-32} . What the inversion cannot undo is the cap at `metagga._ALPHA_MAX = 100`: the capped low-density tail rows, whose exact indicator spans $\sim 10^2$ to $\sim 7 \times 10^6$, carry 100.0 percent of the weighted exchange MSE on the O atom and on H₂O, departing from the exact- τ libxc targets by a median 2.55×10^{-4} and at most 5.70×10^{-4} in enhancement factor. The first reading recorded here – the floor attributed to the smoothed-column/exact- τ asymmetry, with the $\alpha = 0$ mesh nodes reproducing it – priced a computation (1.9×10^{-14}) that the run’s code path never performs, and its agreement in decade with the measured floor was coincidental; it is superseded (HISTORY 2026-08-31, erratum). Nothing is repaired because nothing is broken: the ceiling is the documented energy-faithfulness bound, and the certificates gate its energy consequence $194\times$ inside the 1.0 mHa threshold.

Panel (a) draws p_δ (solid) against the hard clip it replaced (dashed) over the band $|x| \leq 5 \times 10^{-5}$ of raw indicator values: the two are indistinguishable away from the origin and separate only where the clip’s derivative is one-sided,

Smooth positive part at width $w = 1e - 05$

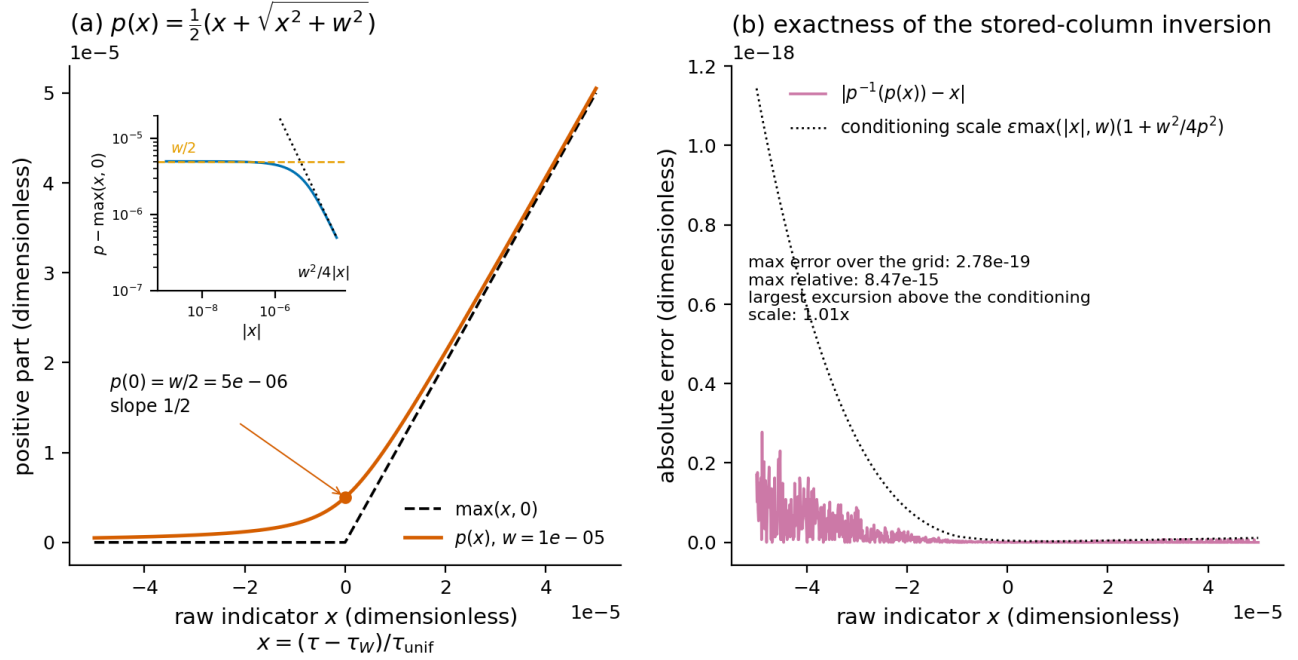


Figure 3: The smooth positive part at width $w = 10^{-5}$ (the δ of this section): (a) p against the hard clip $\max(x, 0)$, with the excess over that clip inset; (b) the absolute error of the round trip through `metagga.invert_smooth_positive_part` against the conditioning scale of the inversion.

the marked value at the origin being the floor $p_\delta(0) = w/2 = 5 \times 10^{-6}$, at which the slope is $1/2$ rather than 0 or 1 . The inset follows the excess $p_\delta - \max(x, 0)$ down from that floor onto its own $w^2/4|x|$ asymptote; at the edge of the band the excess is 4.951×10^{-7} against the asymptote's 5×10^{-7} , so the smoothing is confined to a band of a few widths: the relative distortion of x is 0.99 percent at five widths and falls to 0.01 percent by fifty. Panel (b) is the round trip that keeps a stored indicator column readable: p_δ followed by its exact inverse returns x with an absolute error of at most 2.78×10^{-19} over the grid (8.47×10^{-15} relative), 430 of the 1001 grid points returning bit-exactly, and the largest excursion above the plotted conditioning scale $\varepsilon \max(|x|, w)(1 + w^2/4p^2)$ is $1.01x$ its value. The inversion is therefore exact to the floating-point representation, which is what allows the raw τ to be recovered from a column stored under a recorded width.

5.3 Residual tail response. What the smoothing does not change is the indicator's response amplification in the density tail, which is peaked on a shell rather than a power law: on Li's β channel $\max|\mathrm{d}\alpha_{raw}/\mathrm{d}P|$ climbs from 8.0×10^4 above $2\rho_\beta = 10^{-4}$ to 4.07×10^{11} on the $2\rho_\beta \in [10^{-9}, 10^{-8}]$ band (728 points) and falls to 1.2×10^1 below it – a log-log slope of -0.43 against $2\rho_\beta$, not a power of the density. The outermost radial shell ($\rho_\beta = 1.0 \times 10^{-9}$, i.e. $2\rho_\beta = 2 \times 10^{-9}$; 898 points) carries 2.9×10^{-3} Ha per point in one element of the feature-response Fock term (0.57 Ha over the shell), and a 10^{-14} relative change of the density matrix still moves that channel's virtual-virtual Fock block by 0.37 Ha through the smoothed derivative. The response annihilates the occupied orbital of a one-orbital channel exactly, so the SCF fixed point is unaffected; off-manifold probes are not derivative estimates there (`metagga.py docstring`; `xcquinox/alec/DEFERRED_WORK.md` item 30). Separately, the corrected (exact-derivative) potential destabilizes the plain meta-GGA SCF at long horizons on two tail-dominated configurations: `deep_mgga` on H and LiH oscillates under the production mixer at cycle caps 15–25 (late-step residuals 2.2×10^{-1} and 6.7×10^{-2} Ha against 1.1×10^{-10} and 1.2×10^{-8} for the prior, self-inconsistent assembly; H swings through -0.2747 Ha at cap 15) while every rung-3.5-stacked configuration is stable; production training and evaluation run 3 cycles and never enter the regime, and the recorded remedy path for long-horizon runs is energy-side damping or a DIIS mixer, never a reinstated gradient freeze (HISTORY 2026-08-10).

6. Spin-polarized correlation at $\zeta \rightarrow \pm 1$

Symptom. With the spin-polarized correlation channel enabled, entire training runs went to NaN on pools containing fully polarized species (free H and Li, $\rho_\beta = 0$, $\zeta = \pm 1$) (HISTORY 2026-06-03).

Root cause. Two distinct mechanisms. (i) The PW92 spin interpolation is built from $(1 \pm \zeta)^{4/3}$ terms whose second derivatives pair with the SAME sign,

$$\frac{d^2}{d\zeta^2} (1 \pm \zeta)^{4/3} = \frac{4}{9} (1 \pm \zeta)^{-2/3},$$

so $f''(\zeta)$ diverges at $\zeta = \mp 1$, where the corresponding factor vanishes – a genuine curvature pole at full polarization (Perdew and Wang, Phys. Rev. B 45, 13244 (1992), eqs. (8)-(9)). The full SCF differentiates v_c (itself a first derivative of E_c) a second time, so the exact boundary produces a NaN training gradient on every fully polarized species. The $(\rho_\alpha - \rho_\beta)/\rho$ chain is a separate mechanism, not a pole at the boundary: its second derivative is finite at $|\zeta| = 1$ and diverges only as ρ^{-2} where the total density vanishes, which is what the density floor below guards. (ii) Diffuse-basis tail quadrature noise can drive the total density slightly negative mid-SCF (~ 1300 tail points under a pretrained network at cycle 0; oneshot.py, `_RHO_TOT_FLOOR` comment); a floor of 10^{-300} then squares to 10^{-600} , which underflows to zero in the potential’s forward-mode quotient rule, giving ∞ , and the saturated clip’s zero derivative turns it into $0 \cdot \infty = \text{NaN}$. The energy path is forward-only and stayed finite – hence the discriminating signature of finite energy with a NaN potential (HISTORY 2026-07-04, Phase 17).

Remedy. One shared helper (`oneshot.uks_zeta`) applies, identically in the energy, the per-spin potential, and the feature-derivative accumulation: $\zeta = \text{clip}((\rho_\alpha - \rho_\beta)/\max(\rho, 10^{-12}), -1 + 10^{-6}, 1 - 10^{-6})$, with the gradient frozen on the non-physical $\rho \leq 10^{-12}$ tail. The clip’s forward bias is $O(\epsilon \cdot dE_c/d\zeta) \sim 10^{-8}$ Ha; the invariant that the guards match across paths (so v_c remains the exact gradient of E_c) is held by the single definition and a test rather than by parallel comments (`oneshot.py`).

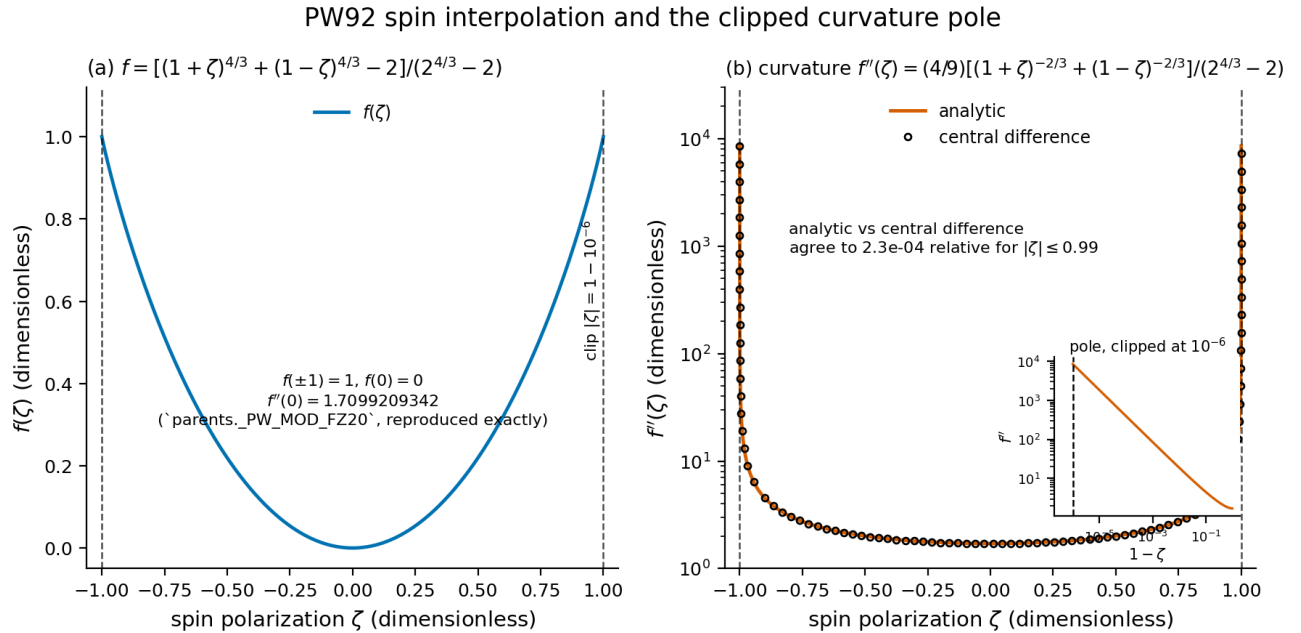


Figure 4: The PW92 spin interpolation $f(\zeta)$ and its curvature: (a) f over the full polarization range with the production clip at $|\zeta| = 1 - 10^{-6}$ marked; (b) f'' on a logarithmic ordinate, analytic against a central difference, with the approach to the pole inset.

Panel (a) shows that f itself is unremarkable at the boundary: it is bounded, rises smoothly to unity at full polarization, and at the clip already reads 0.999996787688811, so excluding the last 10^{-6} of polarization costs 3.2×10^{-6} of the interpolation’s range. Panel (a)’s annotation records the unpolarized curvature 1.7099209342 (`parents._PW_MOD_FZ20`, reproduced exactly). Panel (b) carries the failure: the analytic curvature climbs from

that value to 8550.14 at the clip, a factor of 5000, with the inset showing the $(1 - \zeta)^{-2/3}$ divergence up to the clip, the drawn data terminating on the dashed clip line. The open circles are a central second difference of the written f , tracking the analytic curvature to 2.3×10^{-4} relative for $|\zeta|$ up to 0.99, so the pole drawn here is a property of the interpolation and not of the differencing. The dashed verticals are the clip: outside them lies the region in which the second derivative the full SCF takes of v_c returns a non-finite training gradient, and the clip’s purpose is to keep the quadrature from ever evaluating a point there.

Status. Closed; fail-loud non-finite guards run in every training loop, and the gradient-level sweep described in Sec. 8 closes the one-step blind spot this class originally exploited.

7. C2H2 and the open non-finite-gradient class

History. C2H2 (with HCN and C2H4) drove the earliest degenerate-eigenvalue gradient failure in the project: reverse-mode differentiation through `eigh` at the degenerate π shells produced spurious large gradients, cured by rebuilding the RKS density matrix in occupation-mask form $2(C \odot occ)C^T$ rather than through an eigendecomposition of the density (HISTORY 2026-04-27, Phase 3).

The open class. Two v6 incidents share the finite-loss/non-finite-gradient signature of Sec. 8 and remain open. (i) On the first group (G1, the size ladder), the `medium` subset-size-26 cell failed on the open non-finite-gradient defect, and the published G1 figure set carries 29 of 44 cells with that cell excluded; the record does not name the failing training group of that cell (HISTORY 2026-08-31; the set regenerated at 29 cells on the repaired evaluations, HISTORY 2026-09-01). The twin cell on the attention architecture – `medium_attn` at the same subset size 26 – has since completed and evaluated cleanly, so the defect is specific to the failed cell’s own trajectory rather than a property of the largest subset. (ii) The DM-carrying group’s (G3) preflight compile smoke (run_20260827T163335Z, preflight job 2138042, spec 21 = `deep_combined_attn_3x16` at subset size 26) aborted at per-molecule step 27 on the group `bh76:C2H2` with a FINITE loss of 18.419290403706498 and 36 of 36 gradient leaves non-finite (first leaf `.xnet.net.layers[0].weight`, 80 NaN elements, 0 Inf) (scratch/v6_diag/g3_dm/logs/compile_smoke_probe.out).

Reproduction attempts on the preflight incident. Local replays of the aborting group (`bh76:C2H2` with the injected `anchor:H` regularizer) on the same architecture (`deep_combined_attn_3x16`) at the anchored v6 model class with shape padding returned finite losses (8.5×10^{-6} – 1.4×10^{-4}) and 36/36 finite gradient leaves at `def2-svp/grid 1`, `6-31++G**/grid 3`, and the production identity, from both fresh and pretrained starts and with the density and V_{xc} channels active (scratch/v6_diag/repro_c2h2_grad_attn_svp.log, repro_c2h2_grad_attn_631ppgss_g3.log, repro_c2h2_grad_attn_prod_g3.log, repro_c2h2_grad_attn_prod_g3_pretrained.log, repro_c2h2_grad_V2_channels.log). The defect therefore does not reproduce off-cluster at these identities; a cluster-side decomposition probe is queued, and no root cause is asserted here.

Status. Open at the time of writing; the G1 cell is withheld from the published figures, and recovery of both incidents is deferred until the probe reports.

8. Degenerate-eigenvalue backward amplification: the symmetry-breaking shift

Symptom. A production training replay aborted at step 4 on the group `bh76:OH+N2_to_H+N20` with a FINITE loss (1.085617×10^{-2} , every component finite) and all 36 gradient leaves NaN; the same state evaluated unpadded, in the same process, was finite, with a 0.219% forward-loss shift – orders beyond round-off unless amplified (HISTORY 2026-07-19 to 2026-07-20, Phases 31-32).

Root cause. The `eigh` reverse-mode rule carries $1/(\lambda_i - \lambda_j)$ factors, and the eigenvector backward amplifies matrix-level round-off ϵ by ϵ/gap^2 . The graded symmetry-breaking diagonal that floors symmetry-exact degeneracies was sized at 10^{-8} against forward accumulation noise, ignoring the backward amplification: at 10^{-8} the ratio is $\epsilon/\text{gap}^2 = 2.2$ – order unity – so graph-order round-off differences introduced by the shape-padding pass turned the backward pass into non-finite garbage on the padded graph while the unpadded evaluation of the identical state stayed finite. A five-way single-seam on-cluster ablation isolated the constant: raising `SYM_BREAK_SHIFT` to 10^{-6} alone cured the failing epoch (step-4 loss unchanged to all seven printed digits), the combined-eigh toggle

containing it also cured – a second, independent confirmation – and the leading density-tail candidate and two other seams left the NaN in place (HISTORY 2026-07-20, Phase 33).

Remedy. `SYM_BREAK_SHIFT` = 10^{-6} , centered in the admissible window $[4.7 \times 10^{-7}, 3 \times 10^{-6}]$ whose lower wall is $\epsilon/\text{gap}^2 \leq 10^{-3}$ and whose upper walls are the 3×10^{-5} orientation-lock splitting (which the quasi-random diagonal must not rival) and the Weyl bound (measured maximum eigenvalue motion 9.99×10^{-7}). Reference caches produced under the old constant carry at most $\sim 10^{-6}$ Ha inconsistency. The enabling instrument is the gradient-level abort guard: the loss-only guard had a one-step blind spot (a finite-loss / non-finite-gradient step passes, corrupts every weight, and aborts one step late on the NEXT group, misattributing the origin – the `ae:CO` group was exonerated exactly this way); every training loop now sweeps the gradient pytree before the optimizer update and names the first non-finite parameter path (HISTORY 2026-07-19, Phase 30).

Status. Closed; the constant is pinned by a two-sided sizing test, and the same finite-loss/NaN-gradient detection instrument is what flagged the open Sec. 7 class.

9. Na2 and relative-error denominator collapse

Symptom and cause. Three incidents of one arithmetic class, in which a relative error is formed against a target near zero. (i) Na2’s atomization-energy error was inflated 1340 $\times$ by a relative-AE loss with no floor on the reference magnitude (HISTORY 2026-06-10). (ii) BH76 reactant species (HO, CH3, N2, F2) entered specs with placeholder targets of 0.0, collapsing the relative-error denominator to 10^{-8} so a ~ 0.5 Ha residual blew up to $\sim 2.5 \times 10^7$ (HISTORY 2026-05-10, Phase 6). (iii) A Li+ spec processed after neutral Li overwrote the shared atom anchor index, training the cation toward the neutral anchor – a ~ 5 eV ionization-potential bias (HISTORY 2026-05-10).

Remedy and status. An AE target floor; neutral-only (H, Li) atom anchors; auxiliary reaction species excluded from the AE channel (`aux_only_names`); and the reaction-form AE (`ae_as_reactions`), which scores atomization against the network’s own self-consistent atom energies instead of fixed anchors. One follow-on defect is part of this record: the resolved-config serializer silently dropped `ae_as_reactions`, so three run generations trained the fixed-anchor form their source configuration had turned off; the serializer now round-trips every field by construction (HISTORY 2026-08-10). Closed.

10. H2 and N2: protocol exclusions at the meta-GGA level

The meta-GGA variant of the DFS pretraining protocol drops H2 and N2, giving 28 systems against the GGA level’s 30 (`xcquinox/alec/dfs_pretrain_set.py`, `MGGA_EXCLUDED`). The fidelity-certificate oracle pool restores N2 through its fixed benchmark systems, so a SCAN-parent architecture certifies over 38 systems where a PBE-parent one certifies over 39 – one fewer atomization – with the level selected per parent by `fidelity.dfs_level_for_parent` (`xcquinox/alec/cluster/fidelity.py`; HISTORY 2026-08-31). The five meta-GGA family architectures pass that certificate at the production identity with worst free-atom error 5.2×10^{-3} mHa and worst atomization error 2.5×10^{-3} kcal/mol against the 1.0/1.0 gate (HISTORY 2026-08-31). Closed by construction; recorded because a 38-versus-39 system count in the certificates is designed, not a coverage defect.

11. The geometry-units incident (angstrom read as Bohr)

Symptom and cause. The held-out geometry reader divided already-angstrom coordinates by the Bohr-per-angstrom factor, shrinking every held-out molecule by $\sim 1.89\times$ and garbaging all BH76/W4-11 reaction energies for the networks AND the PBE baseline alike; the corrected BH76 PBE MAE is 11.82 kcal/mol where the corrupted pipeline reported 182 (HISTORY 2026-05-31). Training was never affected (its geometries come from a separate source).

Detection and remedy. Coordinates are read as angstrom and the JSON caches were rebuilt; a Hartree-units guard now checks every frozen reference reaction and ionization energy at loss construction, raising above a 10

Ha sanity ceiling and catching the $\sim 627\times$ kcal/mol-vs-Hartree unit-error class before a loss is built (HISTORY 2026-06-02; xcquinox/alec/losses.py:996-1008). Closed.

12. Per-species OEP overrides

Symptom. The Wu-Yang optimized-effective-potential inversion that generates V_{xc} references converges to the default density-error tolerance for most of the pool but plateaus above it for a small set of species.

Remedy. Plateau detection (objective and density-error trend deque) terminates a stuck inversion rather than burning the iteration budget, and a per-species override table carries bespoke tiers discovered by a tuning sweep: Be, C+, F2, F2O, HF, HS, N2O, and O3 (minimum achieved density errors 4.63×10^{-3} , 1.40×10^{-2} , 9.43×10^{-3} , 4.84×10^{-3} , 4.13×10^{-3} , 1.19×10^{-2} , 4.70×10^{-3} , 9.22×10^{-3} , each accepted at $1.7\times$ its own minimum), plus a manual CF4 entry (plateau 2.486×10^{-3} , just above the 2×10^{-3} RKS default) (xcquinox/alec/external_refs.py, `_PER_SPECIES_OEP_OVERRIDES`; HISTORY 2026-05-06, Phase 6). The intermediates cache is keyed on the grid level so a grid change cannot silently reuse stale references. Closed; the override tiers’ literature annotations remain flagged for pre-publication verification (external_refs.py:838-843).

13. SCF-trajectory instability under trained functionals

Symptom. With the SCF depth raised from 3 to 25 cycles (`full_25`), held-out reaction MAE collapsed to 75–110 kcal/mol against 13–19 for the 3-cycle runs and ~ 15 for PBE, uniformly across matched specs (HISTORY 2026-06-24, Phase 11).

Root cause. Training and evaluation both scored ONLY the final SCF cycle’s energy. For hard species the NN-driven SCF, given 25 cycles, left the stable PBE-initialized basin and the constant linear mixer could not damp it: the captured per-cycle traces show a clean period-2 oscillation in the tail (`t-hooo` steps 18–24 alternate -223.40/-223.75/-223.32/-223.72/-223.31/-223.71/-223.30, a peak-to-peak swing of 0.45 Ha) or a still-drifting endpoint (`s4-c2v` ends 10.3 Ha, ~ 6500 kcal/mol, off PBE). The final-step energy is then an arbitrary oscillation phase. Splitting held-out reactions by SCF convergence proves the mechanism: converged reactions score MAE ~ 12 –28 kcal/mol, non-converged 140–485, with 26–59 non-converged species per spec (HISTORY Phase 11).

Remedy. The DFS trajectory supervision was ported verbatim from the reference implementation: a tail-weighted energy loss (the last $\min(N, 10)$ cycles with quadratically rising weights, generalized to any cycle count) and a step-decaying mixer $\alpha_{mix} = 0.3^{step} + 0.3$, plus tail-weighted reporting. After the fix the 25-cycle run is on par with the 3-cycle one (validation-best held-out median 15.5 against 16.4 kcal/mol, mean 28.07 against 23.69, PBE ~ 14.9) – a partial, in-progress comparison at the time of measurement, the 25-cycle run having completed 23 specs against 73 (HISTORY 2026-07-02).

Related artifacts. (i) The garbage W4-11 atomization energies of the early held-out record were a separate undertraining verdict – SCF divergence of undertrained networks scored verbatim – and not an evaluation bug; three genuine train/held-out overlap leaks were repaired independently of it (HISTORY 2026-06-04). (ii) Genuine three-cycle SCF divergences scored verbatim persist at low rates and track the descriptor stack: first-cycle energy residuals above 0.1 Ha on 6.4% of rows for the multishell rung-3.5 architecture, 3.5% for the attention rung-3.5 form, 0.4–0.9% elsewhere (HISTORY 2026-08-20, Phase 39). Status: closed for the trajectory-supervision class; the per-architecture divergence rates are disclosed on the figures rather than repaired.

14. Bounded-map saturation of the anchored parameterization

Context. No architecture pretrained to its parent functional under the point-wise protocol: on frozen parent densities, the worst-system atomization offsets from the parent ran 25.7 to 56.1 kcal/mol per descriptor-carrying architecture against 4.1–4.2 for the descriptor-free controls (per-system magnitudes overlap at the low end – one descriptor-carrying cell reads 3.5 kcal/mol against a control range of 2.3–4.2; worst values per architecture recomputed from the recorded H2O/N2/CH4 offset table), driven by a shared H-atom pretraining error (+13.7

mHa against +0.8) multiplied by the hydrogen count and by molecular extrapolation of features the atoms-plus-mesh pretraining set never constrained (HISTORY 2026-08-20, Phase 39). The remedy anchors every network to its parent at initialization: the enhancement factor is built as

$$F = 1 + L(z_{parent} + g_\theta), \quad L(x) = \Lambda \sigma(x - \ln(\Lambda - 1)) - 1, \quad z_{parent} = \ln \frac{(\Lambda - 1) F_{parent}}{\Lambda - F_{parent}},$$

with σ the logistic function, Λ the bound of the squash (1.804 for GGA-rung exchange, 1.174 for meta-GGA exchange – the DFS ceiling, networks.py:751 – and 2.0 for the correlation non-negativity map), z_{parent} clamped to ± 40 , and the trainable output g_θ zero-initialized so F equals the parent at initialization to round-off (xcquinox/alec/networks.py _AlecLOB; xcquinox/alec/parents.py lob_preimage). Anchored pretrains then sit 8.7×10^{-7} – 9.2×10^{-6} from the parent in $\max |\Delta F_x|$ where the legacy pretrains sat 0.039–0.090 (HISTORY 2026-08-30, 2026-08-31).

The artifact. The anchored correction inherits the map’s saturation: the trainable term carries $L'(z_{parent})$ as a prefactor, measured 0.45 at $s = 0$ falling to 0.007 by $s = 20$ where the parent exchange approaches its ceiling, with the correlation floor the mirror case. The parameterization therefore suppresses trainability exactly where the parent approaches a bound of $(0, \Lambda)$, and the strongest form occurs at the meta-GGA exchange ceiling: SCAN’s exchange enhancement attains the bound in its single-orbital limit ($F_x = 1.174$ at $\alpha = 0$ as $s \rightarrow 0$), so F_{parent} sits AT $\Lambda = 1.174$, the pre-image clamps at $z_{max} = 40$, and the trainable prefactor is zero in float64 – the map returns the parent to within Λe^{-40} and the network cannot move it there (parents.py, lob_preimage docstring). The measured consequence, on subset-size-18 validation-best cells at $r_s = 2$: exchange corrections of every generation converge to one family (+0.07 to +0.16 bump peaking near $s = 3$), while correlation corrections diverge – the unanchored networks grow a large- s correlation correction (+0.79 to +0.92 for $s \geq 2$, built from a flat ± 0.01 pretrained start) that removes PBE’s systematic barrier bias (BH76 mean signed error –0.20 for the unanchored deep_3x16 at subset size 12, against PBE’s –6.6 to –7.5 kcal/mol), whereas the anchored network’s correction collapses past $s = 2$ (+0.29 at $s = 2$ to +0.01 at $s = 6$) and retains the parent’s bias (–7.75 for the anchored medium; the two signed-error cells are not subset-matched) (HISTORY 2026-08-31; notebooks/analysis/figures_dfs_step7_dfs6311_grid3_v6g1_size_val_best/anchored_vs_unanchored_fx_fc.png, pretrain_fx_fc_delta_all.png, trained_fx_fc_delta_best.png in the same directory). On the first anchored group’s published cells the pattern was W4-11 beaten against the cell’s own PBE anchor in 18 of 18 cells (5.99–12.25 against 13.1–13.6 kcal/mol), the combined pool in 17 of 18, BH76 in 5 of 18 (best 6.51 against 7.73) – the first published state, at 18 cells (HISTORY 2026-08-31). On the complete 29-cell set over the repaired evaluations the pattern holds: W4-11 beaten in 29 of 29 cells (5.99–12.25 against 13.08–13.63 kcal/mol), the combined pool in 28 of 29, BH76 in 10 of 29 (best 6.46 against 7.73), each cell scored on its own strict held-out slice (the run’s per-spec eval_holdout_val_best/test_set.csv, recomputed 2026-09-01). Two scope facts temper the BH76 count. No cell in either arm was trained on a barrier height – the three BH76 training points are GMTKN55-BH76RC reaction energies, a documented deviation from the Letter (dfs_pool.py’s bh76_mode block; approved 2026-05-24) – so the held-out barrier slice is an out-of-quantity extrapolation for anchored and unanchored cells alike. And the own-slice counts are not like-for-like across generations: on the union-excluded common slice (notebooks/analysis/rescore_depth_symmetric.py, strict recipe, 119 of 208 reaction identities), the anchored validation-best cells beat their PBE anchors in 29 of 29 cells on BH76, W4-11 and combined alike.

Status. Open as a design question rather than a defect: the anchor bought four orders of magnitude in pretraining fidelity, and whether it costs the extrapolated barrier-height accuracy – no cell in either arm trains barrier heights; the three BH76 training points are reaction energies – is under controlled test (an anchored run of the identical unanchored architecture is queued; HISTORY 2026-08-31). The bh76_mode='barrier_height' path itself was wired on 2026-09-01 (species = reactants + the staged transition state, trained against the GMTKN55 forward barriers 82.6/8.9/104.8 kcal/mol), so a barrier-supervised arm is now a submission decision rather than a code gap.

15. Cross-cutting reproducibility constraints

Three process-level effects repeatedly masqueraded as, or interacted with, the physics above; each is now pinned.

- **Threaded-BLAS non-associativity and degeneracy.** Reduction order is not associative in floating point, and a degenerate or unconverged system amplifies last-bit differences to observable scale: the OH

component scramble (Sec. 3), the H-atom wander (Sec. 5.1), and the O-atom reference PBE SCF differing by 2.2×10^{-7} Ha between two identical processes above one BLAS thread (HISTORY 2026-08-23, Phase 43). Bitwise claims are made only under pinned thread counts.

- **Process-memory-dependent quadrature.** PySCF sizes its XC grid loop, its incore-versus-direct integral path, and its density-fitting block sums from the memory the process has left, so the same specification returned energies and densities differing at the 10^{-13} level with process history (the def2-svp O atom: 1 against 54 grid blocks, the production-basis O atom spanning two; C5H8: 4, 8, or 222 auxiliary blocks at 888 auxiliary functions; methane at 288 auxiliary functions: two different Hartree-Fock energies across memory histories). Every reference mean field now pins the grid block size (12544 points), the integral path (incore iff $n_{ao} \leq 211$), and a fixed 240-vector auxiliary blocking, after which records are bitwise equal across processes holding 0.7 GB to 3.6 GiB at one thread (HISTORY 2026-08-24, Phase 43). A reference whose quadrature order depends on process memory is not a reference; the pins enter no cache identity, so existing caches stay valid.
- **The returned-density convergence ceiling.** PySCF’s SCF converges on the gradient of one Fock form, then runs one extra diagonalization and returns THAT density, accepted under a looser rule – so the density a record carries is one step past the converged point, bounded by $3\times$ the gradient bar rather than the bar. The singlet-CH2 SCAN record rebuilt its orbital gradient at 3.237×10^{-5} against the 3.16×10^{-5} bar on a record stamped converged after 7 DIIS cycles (a bent-CH2 PBE control reaches $2.26\times$ the bar). The generator therefore holds a record first to its own stamped gradient at 10^{-6} relative (record integrity: a density and Fock pieces from different SCFs are refused whatever their gradient) and then to PySCF’s own $3\times$ ceiling (pretrain_data_gen._GRADIENT_CHECK_MARGIN = 3.0), rather than to a bar PySCF does not hold its returned density to (HISTORY 2026-08-25; data.py _REFERENCE_SCF_CONV_TOL comment).

16. Summary table

Species / system	Artifact	Magnitude	Detection	Remedy
C2	Two SCF configurations; non-convergent oscillating DIIS; aufbau flip on dm0 ingestion	50.10 kcal/mol inter-branch gap; 0.12 Ha DIIS spread	Branch tolerance (10^{-4} Ha) against trajectory minimum; cross-spec E_{PBE} spread $> 10^{-4}$ Ha	Orbital-pair-seeded second-order rescue; refusal over silent recording
C2 (reference cache)	Trajectory-dependent grid pruning	26840 vs 26568 grid points	Loader shape gate	First-attempt grid pinned; DF identity stamped; intermediates deleted with the final file
Li	SCAN/diffuse-basis DIIS basin escape	Endpoint 3.4 Ha above the solution	Two-stage non-convergence refusal	Second-order solve from the best-gradient trajectory point (102 cycles total)
Li (β), H	One-orbital channel: $\tau = \tau_W$, indicator a rounding residue; hard-clip derivative	0.93 Ha Fock response under a 10^{-14} density change (0.37 Ha residual after smoothing)	10^{-14} density-matrix perturbation probe of the Fock response	Smooth positive part, $\delta = 10^{-5} \tau_{unif}$; cost 1.2×10^{-7} Ha
Li 2s tail (meta-GGA)	Iso-orbital indicator ill-conditioning at $\rho \sim 10^{-13}$	$d\alpha/d\sigma \sim 10^{59}$; kernel 4×10^{43}	Step-0 NaN loss	Value cap at 100 (smoothed domain $[5 \times 10^{-6}, 100]$); exact feature-response term (no gradient freeze)
H, Li (free atoms)	$f''(\zeta) \rightarrow \infty$ at $\zeta = \pm 1$; negative-tail underflow in the potential JVP	All-NaN training runs	Finite energy with NaN potential	ζ clip $\pm(1 - 10^{-6})$; ρ floor 10^{-12} ; shared guard helper
OH, CH, NO	$^2\Pi$ orientation degeneracy; reference/SCF component mismatch	Up to $23\times$ model-free density-error spread; CH $\epsilon_n = 0.155$ vs median 0.0084	Model-free error varying at constant energy	Traceless-quadrupole h_{core} lock (3×10^{-5}); per-species density weighting recorded
O, N, F (atoms)	Degenerate p-shell orientation; two-block spin-scaling defect	0.1 mHa SCAN E_{xc} spread; 30.1 kcal/mol on O (two-block)	Cross-process record spread; libxc spin=1 oracle	Orientation-locked references; per-channel doubled-density descriptor blocks
C2H2 (+ HCN, C2H4)	Degenerate π eigh gradients (closed); open non-finite-gradient class	G1 cell lost (group unnamed); G3 preflight abort on bh76:C2H2 at finite loss 18.42	Gradient-leaf sweep before the optimizer update	Occupation-mask DM rebuild; cluster probe queued (open)
BH76 OH+N2 group	eigh backward amplification ϵ/gap^2	All 36 gradient leaves NaN at finite loss	Gradient sweep; single-seam ablation	SYM_BREAK_SHIFT $10^{-8} \rightarrow 10^{-6}$ (window $[4.7 \times 10^{-7}, 3 \times 10^{-6}]$)

Species / system	Artifact	Magnitude	Detection	Remedy
Na2 (+ aux species, Li+)	Relative-error denominator collapse	$1340\times$ AE error; 2.5×10^7 residual; ~ 5 eV IP bias	Loss magnitude	AE floor; neutral anchors; aux exclusion; reaction-form AE
H2, N2	Meta-GGA protocol exclusion	28 vs 30 pretraining systems	Certificate system count	Oracle pool restores N2 (38 vs 39 certified systems)
All held-out	Angstrom read as Bohr ($1.89\times$ shrinkage)	BH76 PBE MAE 182 vs 11.82 kcal/mol	> 10 Ha ceiling on frozen reference reaction/IP energies at loss construction	Unit fix; caches rebuilt
Be, C+, F2, F2O, HF, HS, N2O, O3, CF4 t-hooo, s4-c2v	OEP plateau above default tolerance Trained-functional 25-cycle SCF oscillation/drift	Density-error plateaus 2.5×10^{-3} – 1.4×10^{-2} Period-2, 0.45 Ha peak-to-peak; 10.3 Ha drift	Plateau detection Per-cycle energy trace; convergence-split MAE	Per-species override tiers ($1.7\times$ own minimum) DFS tail-weighted loss + decaying mixer
CH2 (singlet)	Returned density one step past convergence	Rebuilt gradient $1.02\times$ the bar on a stamped-converged record	Generator gradient check	Gradient stamp; integrity at 10^{-6} relative; $3\times$ ceiling
Anchored networks	Bounded-map pre-image saturation	L' : 0.45 at $s = 0$ to 0.007 at $s = 20$; BH76 signed error -7.75 vs -0.20	F_x/F_c difference curves vs parent	Controlled anchored-vs-unanchored test in progress